Frozen V3/V4 Descriptor and Probabilistic World-Model Validation

Strictly offline local audit

2026-07-12

1 Decision

Decision C - descriptor semantics incomplete or inconsistent is the single main decision. Strict spatial extrapolation and topology-mask support are inadequate for an EIG pilot.

2 Scope

No exact solver, Delta-q search, active-learning loop, HPC job, or frozen-data edit was performed. Historical learned state is non-transferable.

3 Method

Inputs are $(k_B T/t, J_A/t, \mu/t)$. Outputs use fixed physical transforms and field-specific applicability/reliability masks. Normal-state q is not applicable, never a valid zero. Spatial, slice, corner, future, cross-run, probe-cloud, sequential replay, and retrospective tests were run.

4 Results

See the machine-readable CSV companions for per-field MAE, RMSE, R^2, NLL, interval coverage, uncertainty-error correlation, and replay/control values.

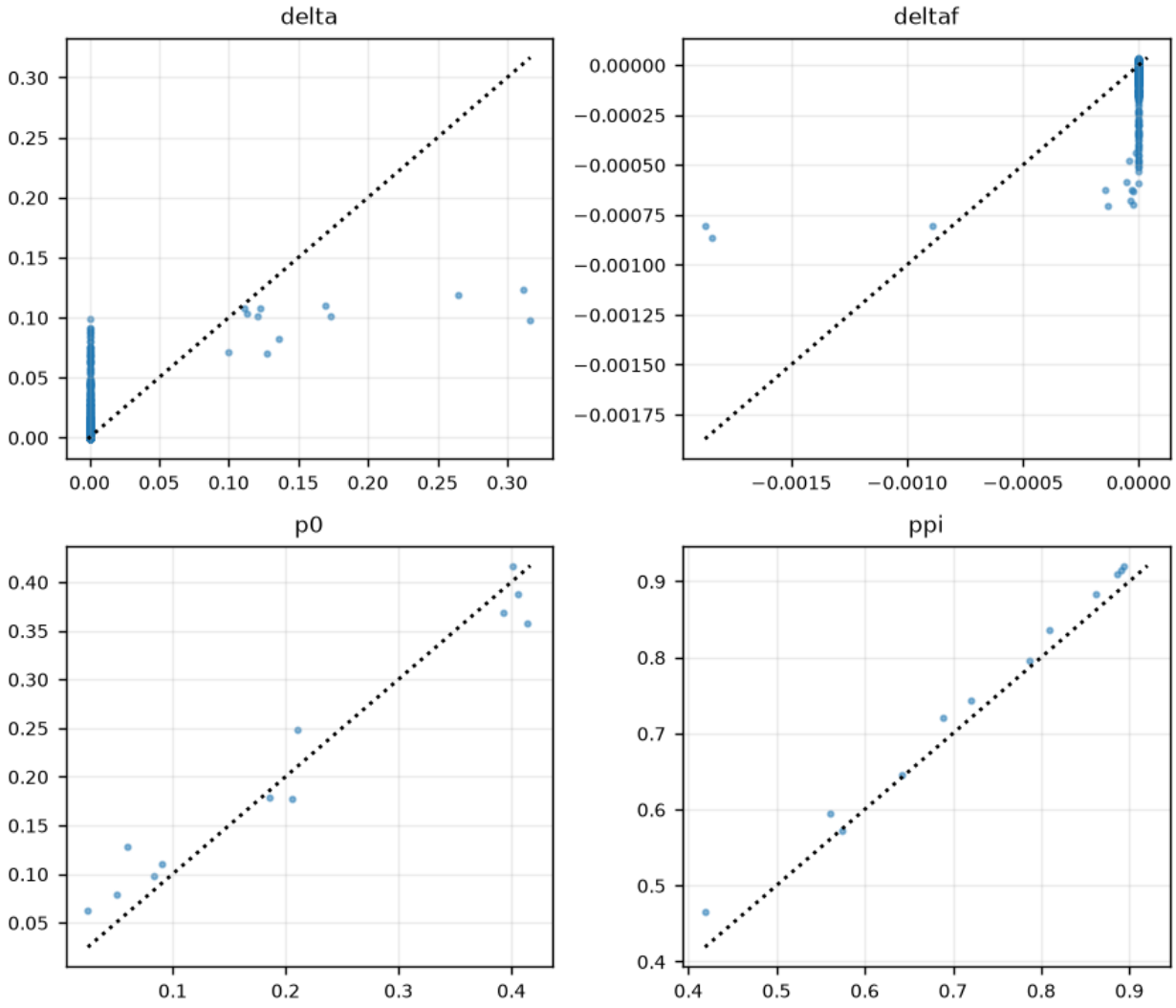


Figure 1: figure 01 predicted exact

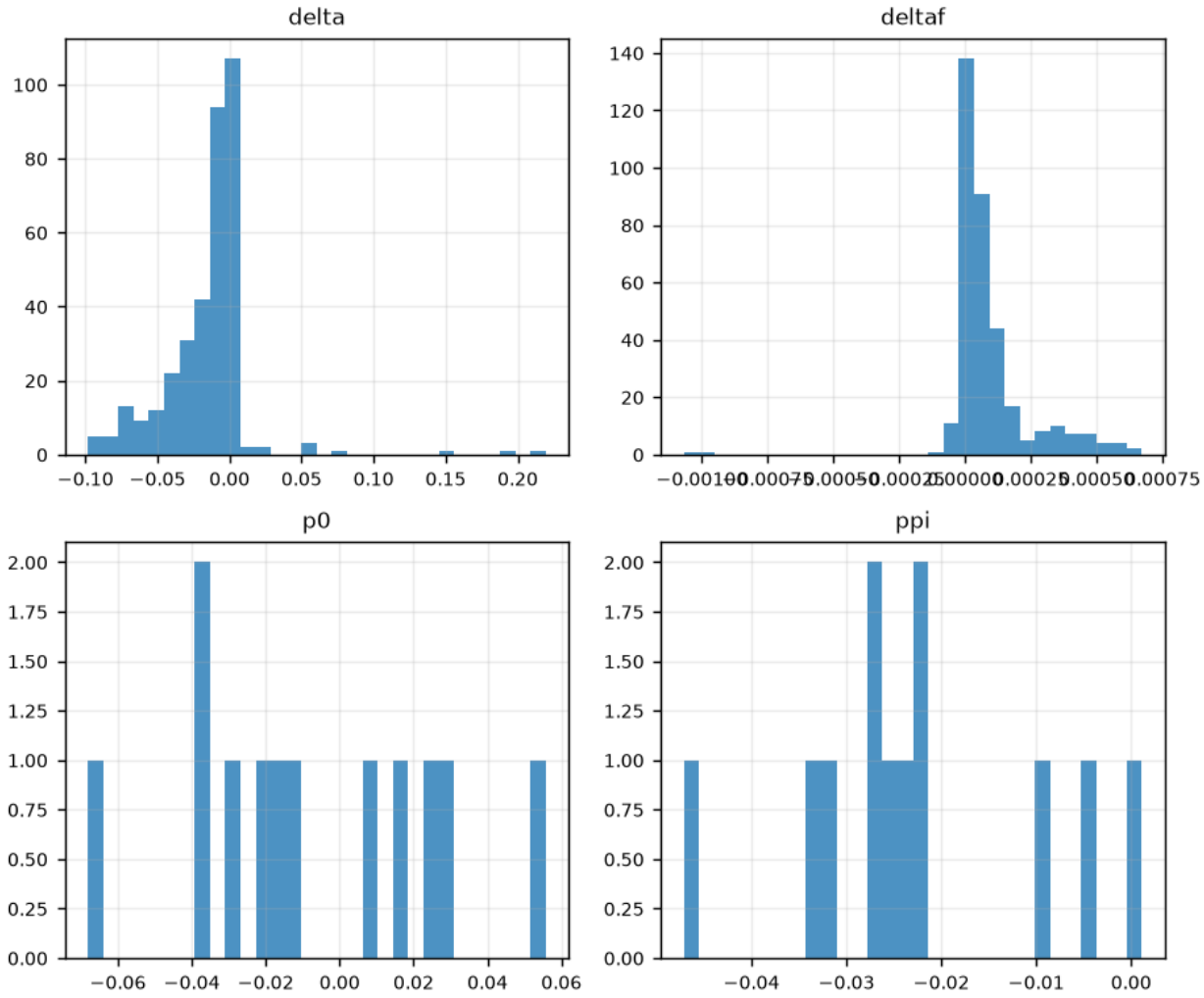


Figure 2: figure 02 residuals

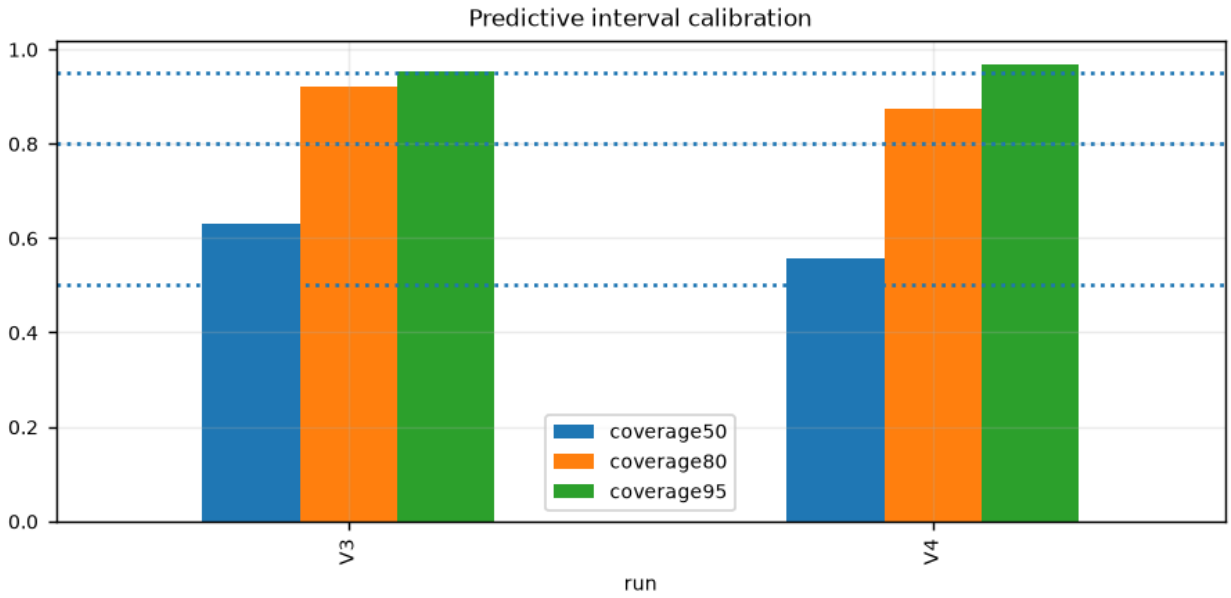


Figure 3: figure 03 calibration

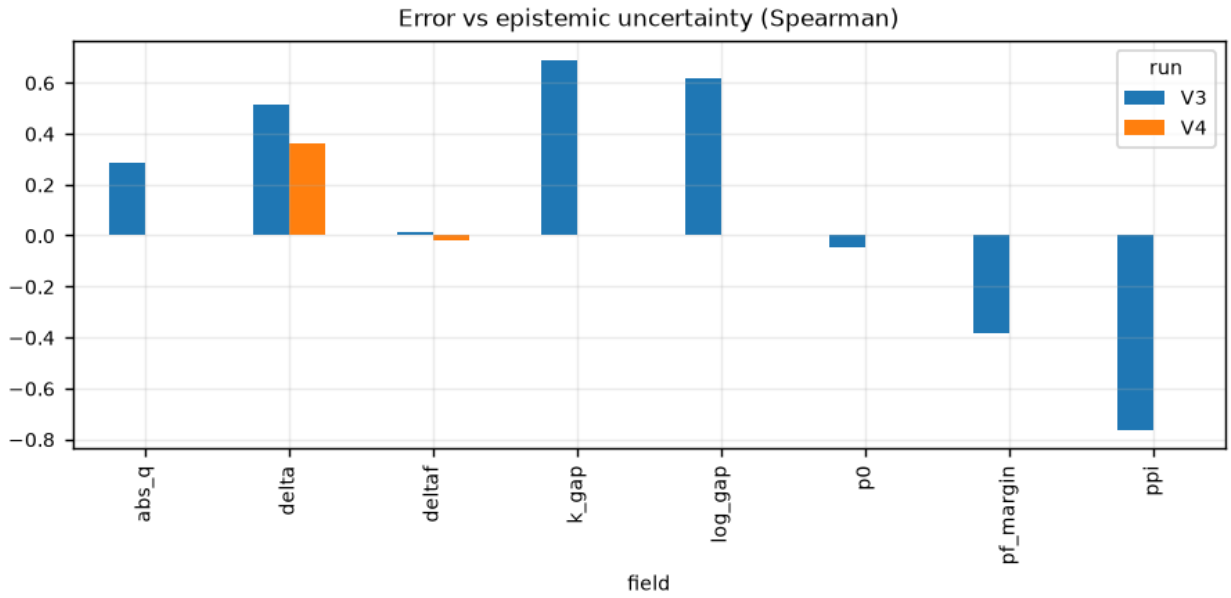


Figure 4: figure 04 error epistemic

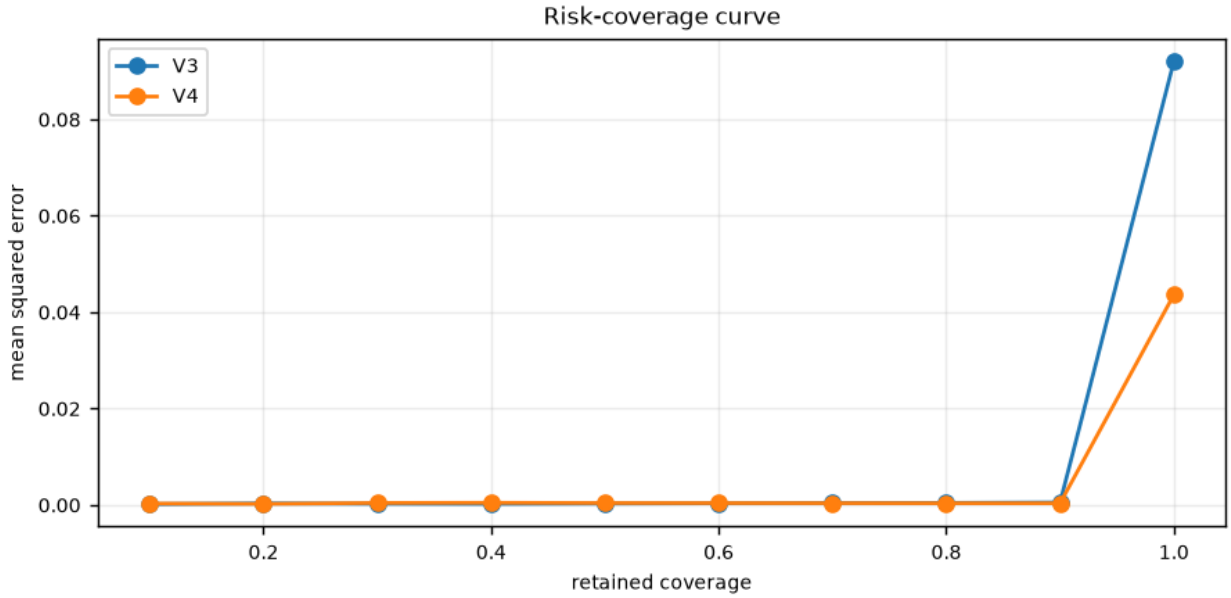


Figure 5: figure 05 risk coverage

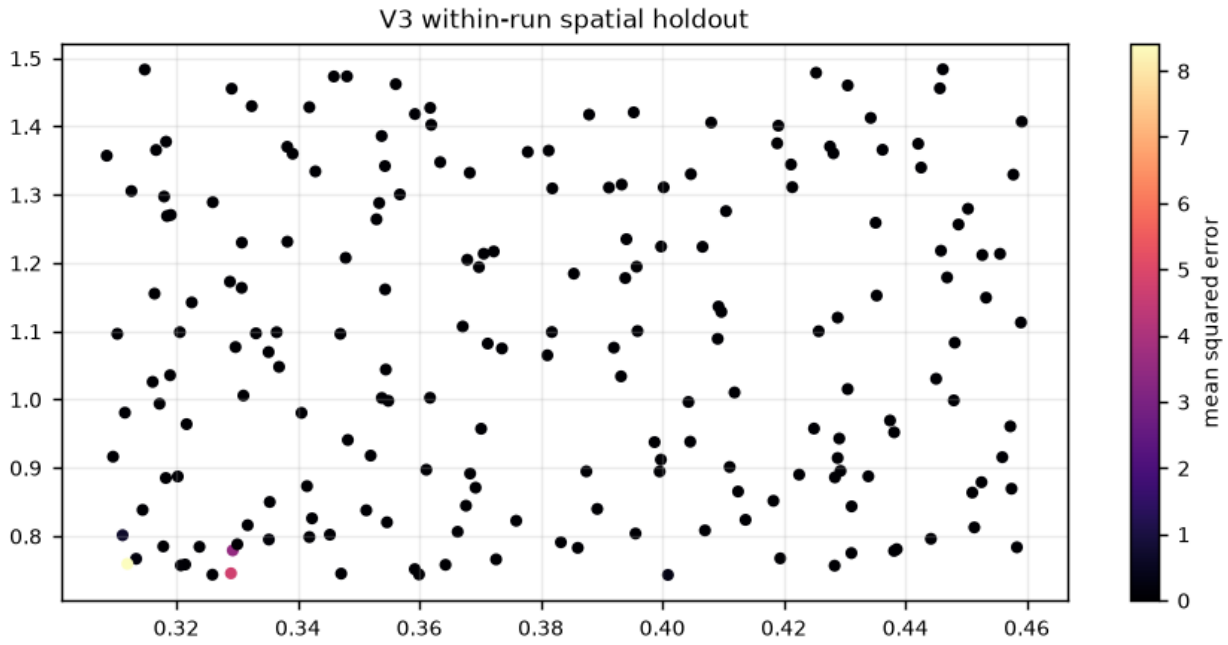


Figure 6: figure 06 v3 spatial

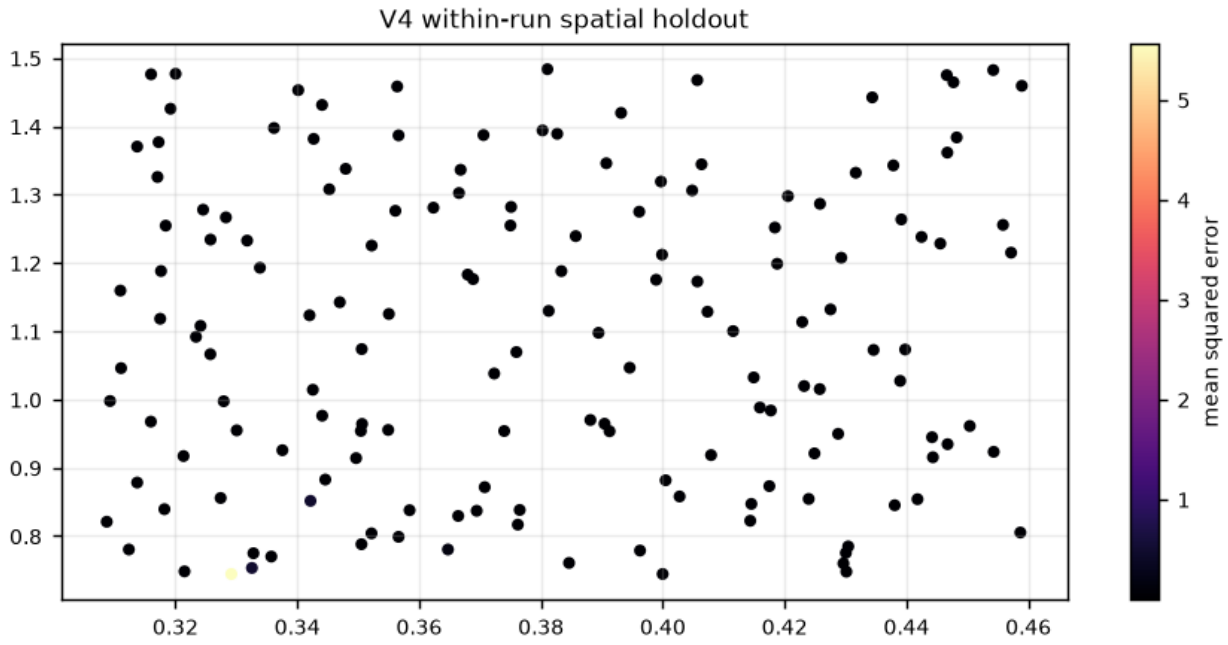


Figure 7: figure 07 v4 spatial

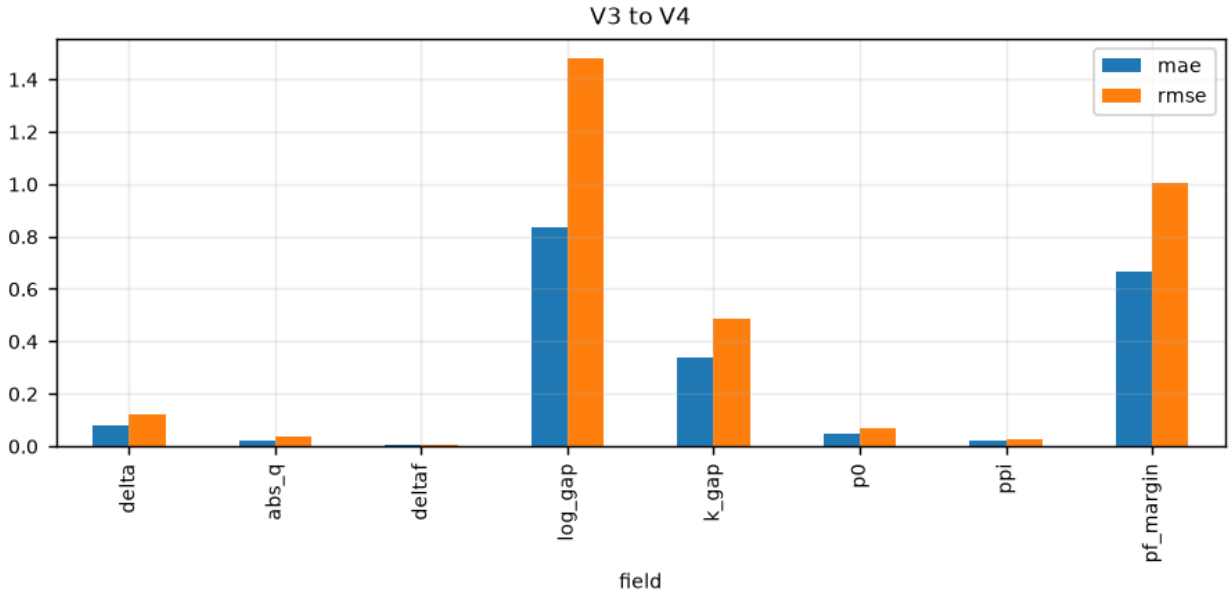


Figure 8: figure 08 v3 to v4

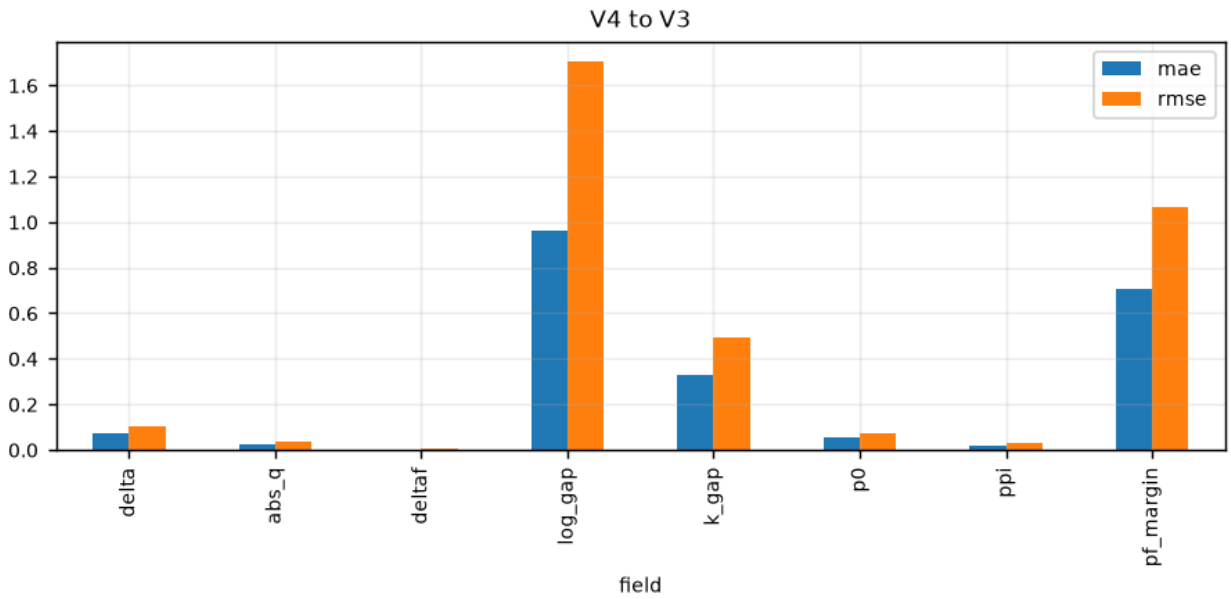


Figure 9: figure 09 v4 to v3

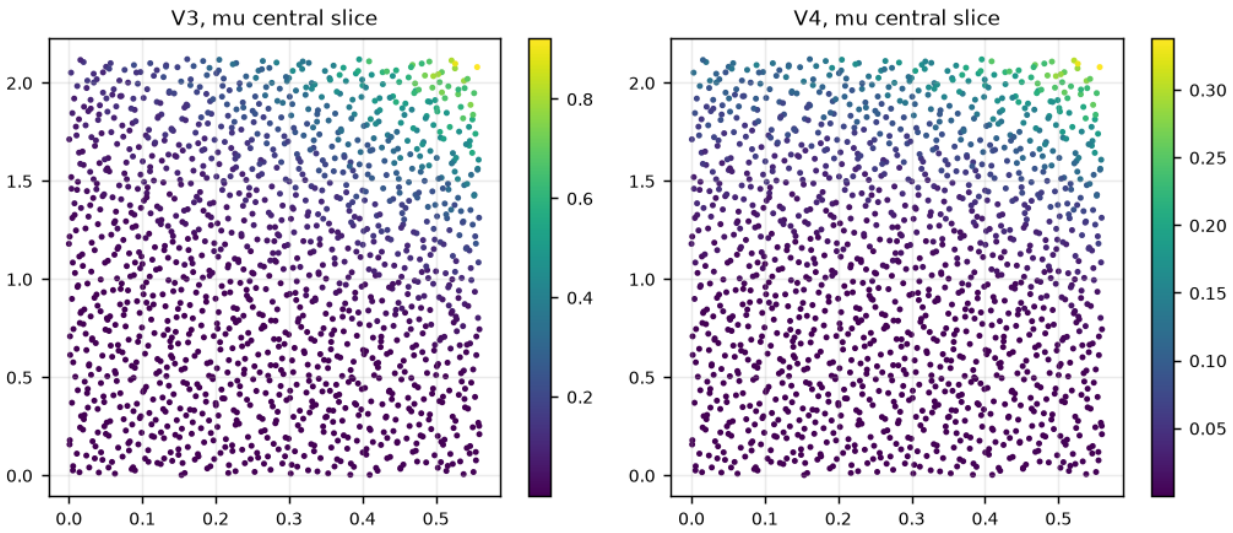


Figure 10: figure 10 probe uncertainty

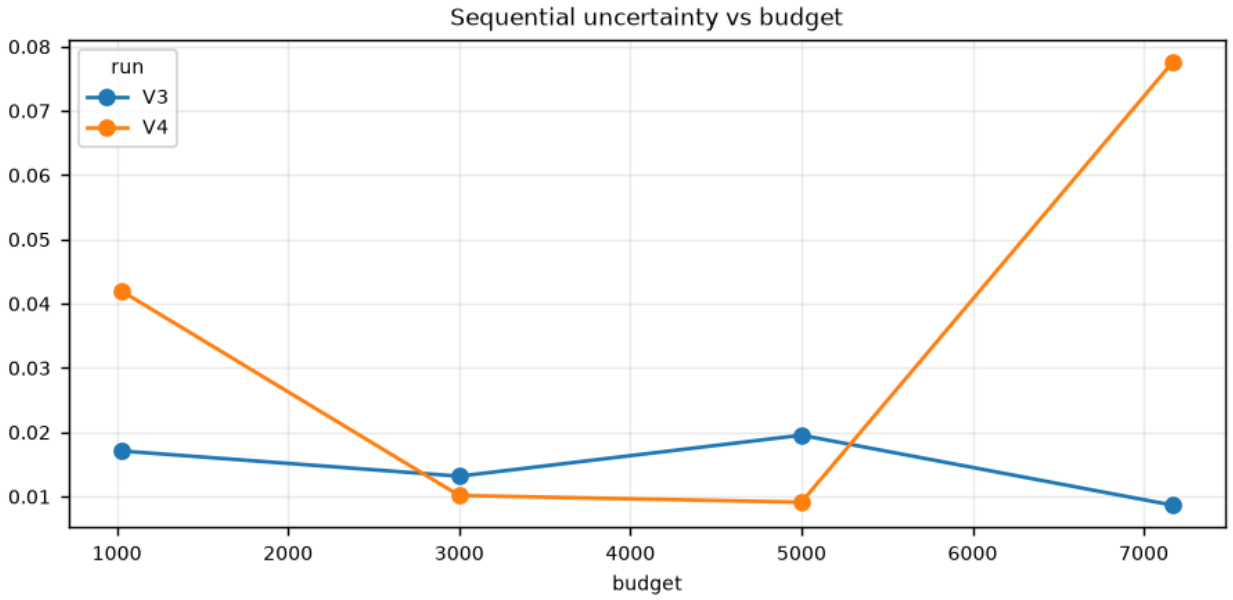


Figure 11: figure 11 sequential uncertainty

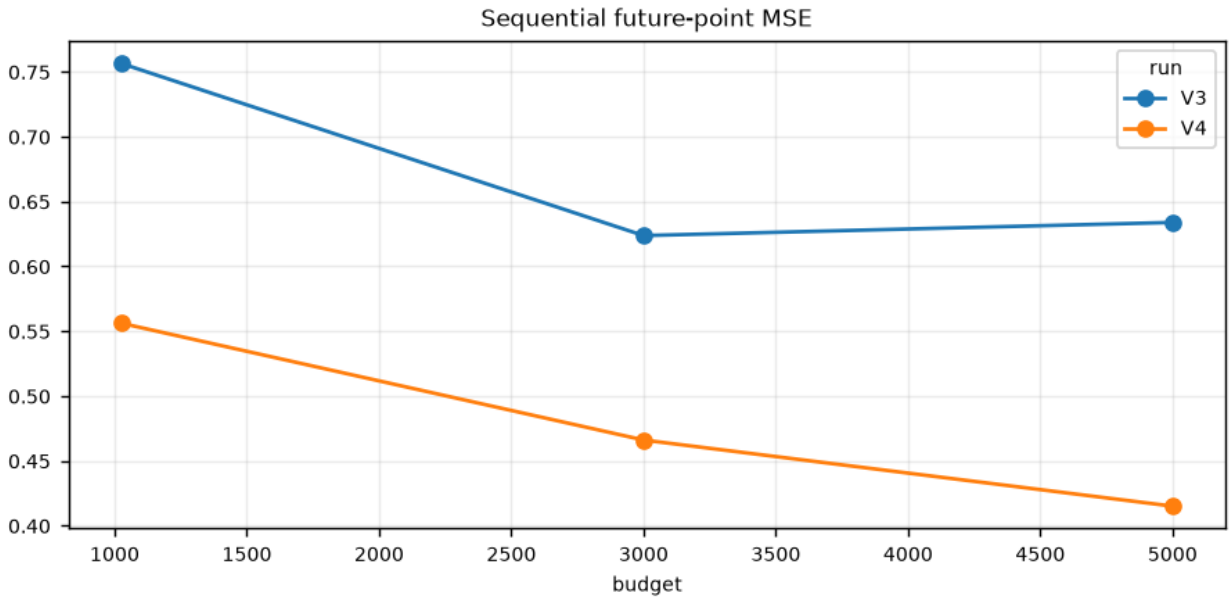


Figure 12: figure 12 sequential future error

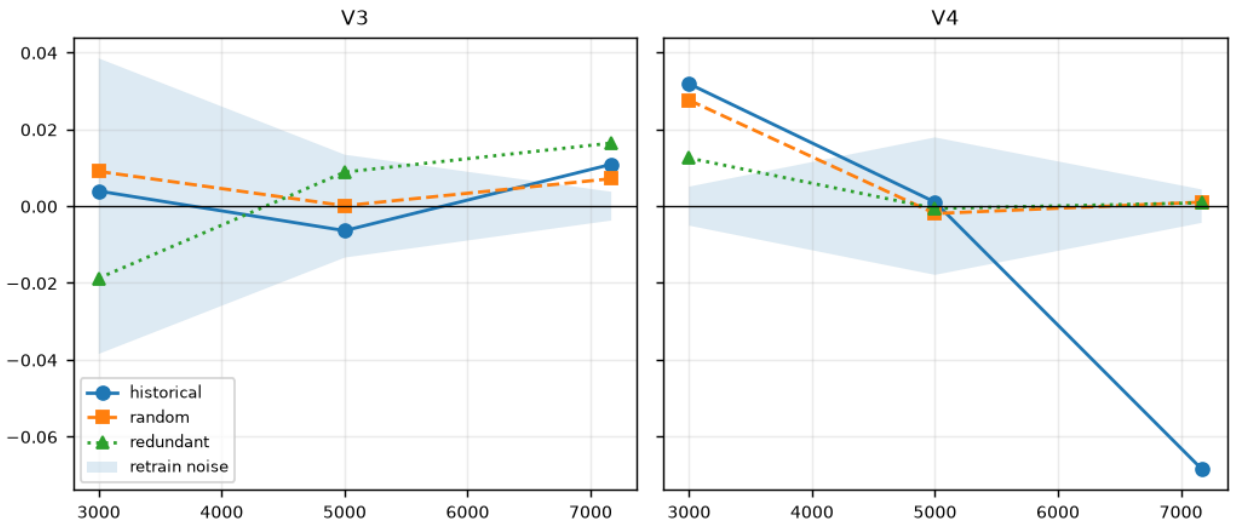


Figure 13: figure 13 retrospective information

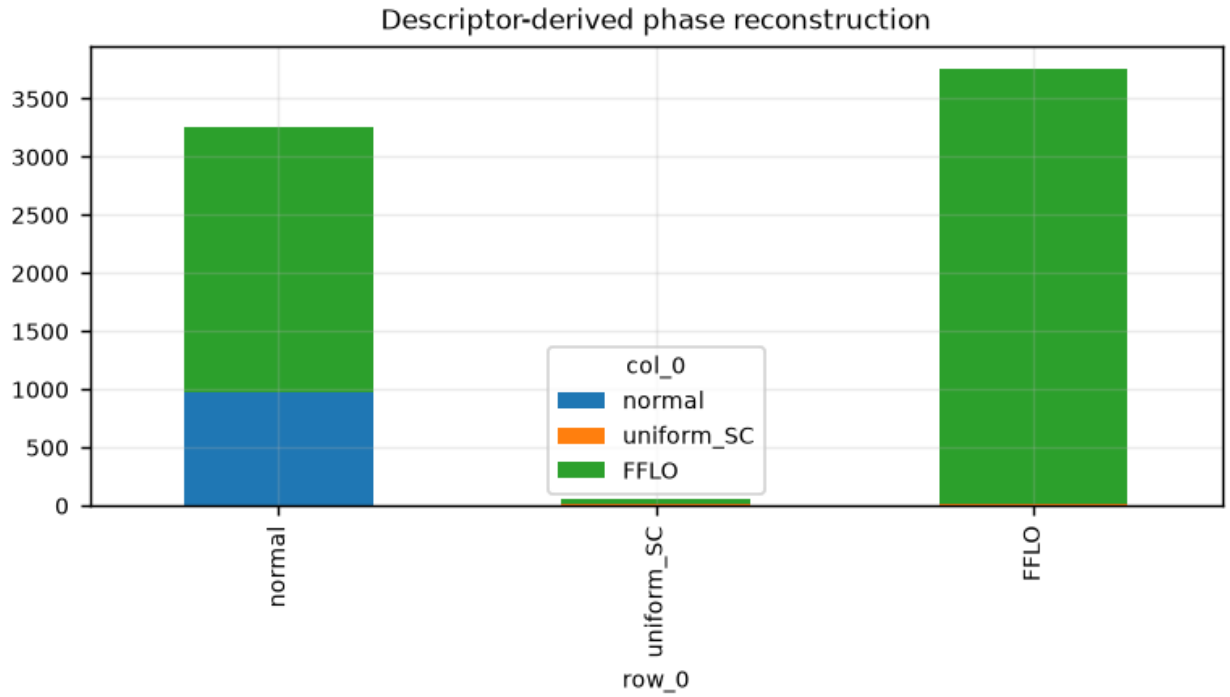


Figure 14: figure 14 phase reconstruction

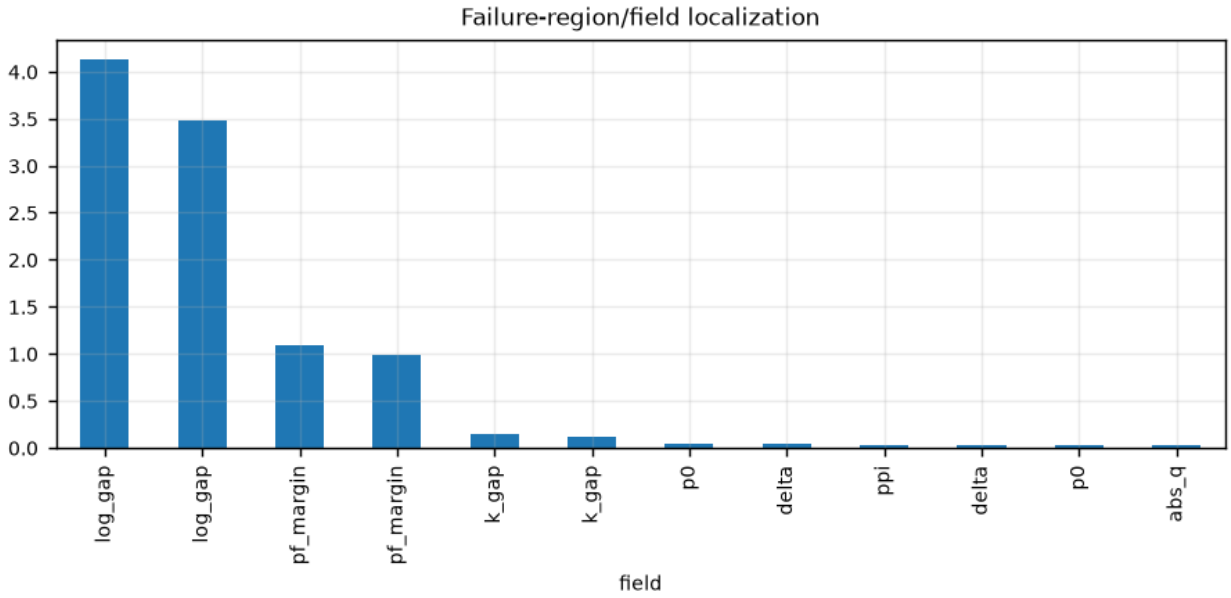


Figure 15: figure 15 failure localization

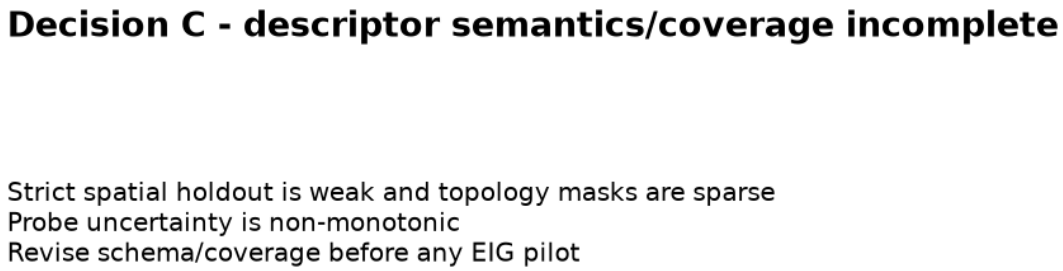


Figure 16: figure 16 decision summary

5 Production isolation

Only schemas, fixed transforms, architecture, loss source, and tests are allowlisted. Data rows, scalers, checkpoints, weights, optimizer state, latent/cluster state, coordinates, rewards, and replay buffers are denied.